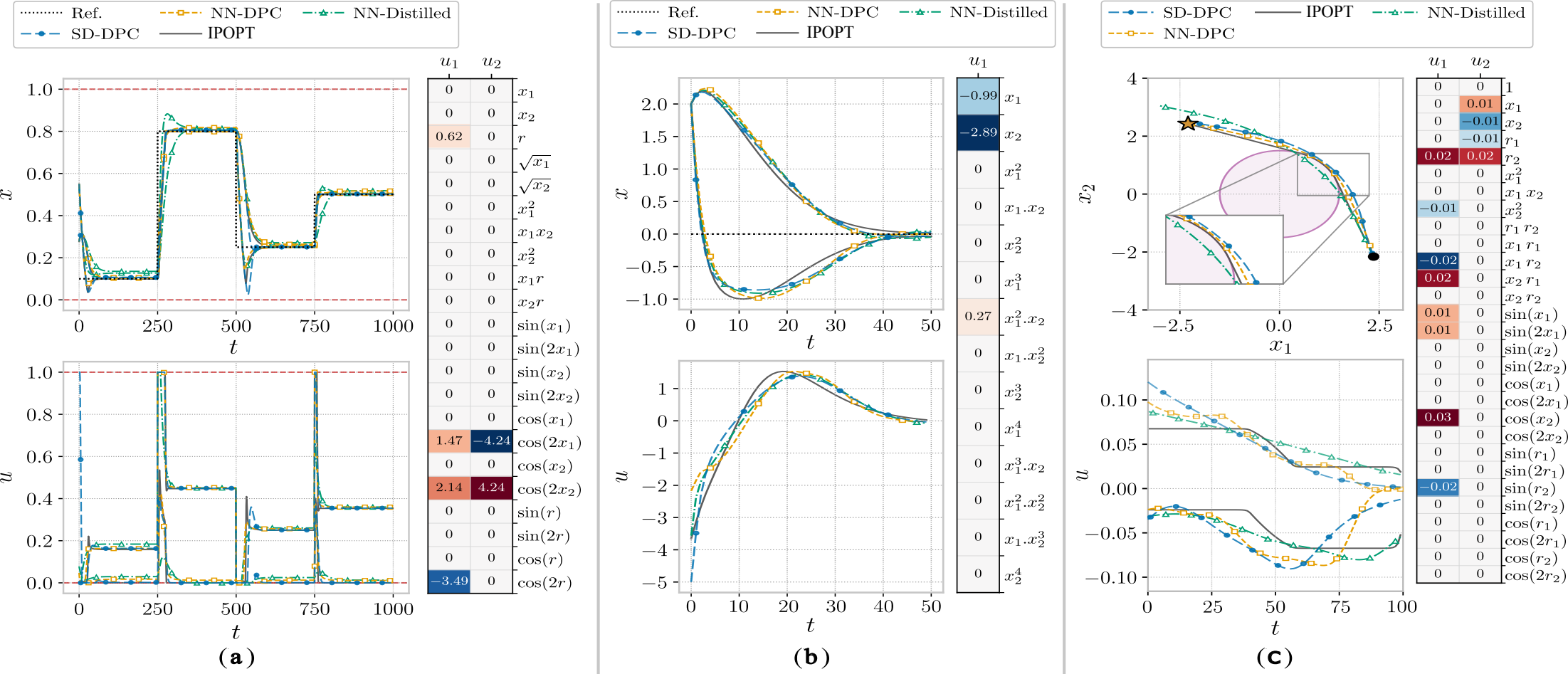

Ref.
SD-DPC
NN-DPC
IPOPT
NN-Distilled
x
u
t
u_1 u_2
0 0 x_1
0 0 x_2
0.62 0 r
0 0 $\sqrt{x_1}$
0 0 $\sqrt{x_2}$
0 0 x_1^2
0 0 x_1x_2
0 0 x_2^2
0 0 x_1r
0 0 x_2r
0 0 $\sin(x_1)$
0 0 $\sin(2x_1)$
0 0 $\sin(x_2)$
0 0 $\sin(2x_2)$
0 0 $\cos(x_1)$
1.47 −4.24 $\cos(2x_1)$
0 0 $\cos(x_2)$
2.14 4.24 $\cos(2x_2)$
0 0 $\sin(r)$
0 0 $\sin(2r)$
0 0 $\cos(r)$
−3.49 0 $\cos(2r)$
(a)
u_1
−0.99 x_1
−2.89 x_2
0 x_1^2
0 $x_1.x_2$
0 x_2^2
0 x_1^3
0.27 $x_1^2.x_2$
0 $x_1.x_2^2$
0 x_2^3
0 x_1^4
0 $x_1^3.x_2$
0 $x_1^2.x_2^2$
0 $x_1.x_2^3$
0 x_2^4
(b)
SD-DPC
NN-DPC
IPOPT
NN-Distilled
x_2
x_1
u_1 u_2
0 0 1
0 0.01 x_1
0 −0.01 x_2
0 −0.01 r_1
0.02 0.02 r_2
0 0 x_1^2
0 0 $x_1\ x_2$
−0.01 0 x_2^2
0 0 $r_1\ r_2$
0 0 $x_1\ r_1$
−0.02 0 $x_1\ r_2$
0.02 0 $x_2\ r_1$
0 0 $x_2\ r_2$
0.01 0 $\sin(x_1)$
0.01 0 $\sin(2x_1)$
0 0 $\sin(x_2)$
0 0 $\sin(2x_2)$
0 0 $\cos(x_1)$
0 0 $\cos(2x_1)$
0.03 0 $\cos(x_2)$
0 0 $\cos(2x_2)$
0 0 $\sin(r_1)$
0 0 $\sin(2r_1)$
−0.02 0 $\sin(r_2)$
0 0 $\sin(2r_2)$
0 0 $\cos(r_1)$
0 0 $\cos(2r_1)$
0 0 $\cos(r_2)$
0 0 $\cos(2r_2)$
(c)